

SOLUTION ARCHITECTURE

Medical Diagnosis Center by ServiceNow Service Portal

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The solution architecture connects the Medical Diagnosis Center problem with the ServiceNow components actually represented in the project's screenshot-based implementation. It shows how a patient moves from diagnostic-test discovery and booking through patient details, scheduling, the configured payment step, appointment tracking, appointment completion, automated report creation and lab-report access.

1. SOLUTION ARCHITECTURE

- Provide a centralized patient-facing diagnostic-service journey through ServiceNow Service Portal.
- Connect diagnostic-test discovery, booking, patient details, scheduling and payment into one appointment flow.
- Maintain structured Patient, Diagnostic Test, Appointment and Report records.
- Automate report creation when the configured appointment-completion condition is reached.
- Provide patient-facing appointment and lab-report visibility through dedicated portal components.
- Maintain controlled access and traceability without claiming autonomous clinical diagnosis or treatment.

2. SOLUTION ARCHITECTURE DIAGRAM

USER / PORTAL LAYER	SERVICENOW APPLICATION / DATA LAYER	AUTOMATION / OUTPUT LAYER
Patient / Service User	Medical Test Catalog + Diagnostic Test	Select Test / Book Now
ServiceNow Service Portal	Patient + Appointment Booking	Patient Details → Schedule
My Appointments	Appointment Record	Appointment Status / Completion
My Lab Reports	Report Record	Auto Create Report on Appointment Complete
Administrator / Diagnostic Staff	ServiceNow Tables / Configuration	Report & Appointment Maintenance
	Roles & ACLs	Access Control / Traceability

Figure 1. Solution Architecture – Medical Diagnosis Center by ServiceNow Service Portal

3. ARCHITECTURE COMPONENTS AND RESPONSIBILITIES

S.No.	Component	Architecture Role	Responsibility
1	Patient / Service User	Primary user layer	Browses diagnostic tests, selects a test, enters patient details, chooses a schedule, completes the configured payment step, views appointments and accesses available lab reports.
2	ServiceNow Service Portal	Patient-facing access layer	Provides the Medical Test Catalog, Appointment Booking, My Appointments and My Lab Reports experiences shown in the project.
3	Medical Test Catalog	Diagnostic-service discovery layer	Displays available diagnostic tests and provides the Book Now entry point for a selected test.
4	Diagnostic Test	Master data layer	Stores the diagnostic-test information used by the Medical Test Catalog and booking journey.
5	Patient	Patient data layer	Stores the patient information used during the diagnostic appointment process.
6	Appointment Booking	Booking transaction layer	Guides the patient through Patient Details → Schedule → Payment → Done and creates the appointment context.
7	Appointment	Operational transaction layer	Stores the booked appointment and its lifecycle/status, and supports the My Appointments view.
8	My Appointments	Patient tracking layer	Displays the patient's booked appointments and their current state.
9	Appointment Completion Automation	Automation layer	When the appointment reaches the configured completed condition, triggers creation of the corresponding Report record.
10	Report	Laboratory-report data layer	Stores the report record created/maintained by the configured report workflow.
11	My Lab Reports	Patient report-access layer	Provides the patient-facing view of available lab report records.
12	Roles / ACLs	Security layer	Controls access to patient, appointment, diagnostic-test and report information according to the implemented ServiceNow security configuration.
13	ServiceNow Automation / Business Rule	Lifecycle automation layer	Connects appointment completion to report creation; exact trigger/action names should match the live implementation.
14	Administrator / Diagnostic Staff	Operations layer	Maintains diagnostic tests, appointments and reports and manages the configured ServiceNow application components.

4. END-TO-END DATA AND CONTROL FLOW

1. Patient opens the Medical Diagnosis Center Service Portal.
2. Patient enters the Medical Test Catalog and browses the available diagnostic tests.
3. Patient selects the required Diagnostic Test and clicks Book Now.
4. Appointment Booking opens the Patient Details stage and captures the required patient information.
5. Patient proceeds to Schedule and selects the configured appointment date/time option.
6. Patient proceeds to the Payment stage and completes the configured payment step.
7. The booking reaches Done and the corresponding Appointment record is created/available in the system.
8. Patient opens My Appointments to view the booked appointment and its current status.
9. When the appointment is marked Completed, the configured appointment-completion automation executes.
10. The automation creates the corresponding Report record for the completed appointment.
11. Patient opens My Lab Reports and views the available report record.
12. Administrators/authorized staff use the underlying Patient, Diagnostic Test, Appointment and Report records to maintain and support the service.

5. MASTER IMPLEMENTED WORKFLOW

Medical Test Catalog → Select Diagnostic Test → Book Now → Patient Details → Schedule → Payment → Done → Appointment Record → My Appointments → Appointment Completed → Auto Create Report on Appointment Complete → Report Record → My Lab Reports

6. DATA MODEL / RECORD RELATIONSHIPS

Record / Object	Role in Architecture	Relationship / Use
Patient	Stores patient information used by the booking journey	Patient information is used when creating/processing the Appointment.
Diagnostic Test	Stores the diagnostic test offered through the catalog	Selected Diagnostic Test drives the Book Now journey and appointment context.
Appointment	Stores the booked appointment and status	Created/maintained through the booking flow and displayed in My Appointments.
Report	Stores the report record	Created/maintained after the configured Appointment Completed event and displayed in My Lab Reports.

7. ARCHITECTURE DESIGN PRINCIPLES

Principle	Application in this Project
Patient-first service flow	The architecture follows the actual portal journey from diagnostic-test discovery through appointment and report access.
Separation of concerns	Catalog, patient data, appointment transaction, automation, report data and portal views are treated as distinct responsibilities.
Centralized service data	Patient, Diagnostic Test, Appointment and Report information is stored in structured ServiceNow records.
Lifecycle automation	Appointment completion is used as the transition point for the configured report-creation automation.
Controlled access	Roles and ACLs protect patient and operational data according to the implemented access model.
Traceability	A patient's journey can be followed from selected diagnostic test to appointment and then to the corresponding report record.
Configuration-first maintainability	The architecture uses ServiceNow portal/configuration and automation components; unsupported external integrations are not assumed.

8. SECURITY AND ACCESS CONTROL

Roles and ACLs should control access to patient, diagnostic-test, appointment and report information according to the configured ServiceNow security model. The architecture separates the patient-facing portal experience from administrative and operational maintenance. Only the access behaviour actually configured in the project should be demonstrated. No additional clinical security model is assumed beyond the implemented ServiceNow roles and ACLs.

9. OBSERVABILITY AND VALIDATION

- Verify that the Medical Test Catalog displays the configured diagnostic tests.
- Verify that Book Now opens the intended booking journey.
- Verify that Patient Details, Schedule and Payment values are captured as configured.
- Verify that Done results in the expected Appointment record/state.
- Verify that My Appointments displays the appointment.
- Complete an appointment and verify that the configured automation creates the Report record.
- Verify that My Lab Reports displays the resulting report record.
- Use the relevant ServiceNow record/configuration views to troubleshoot mapping, status and automation issues.

10. CURRENT BOUNDARY AND FUTURE EXPANSION

- Current architecture is limited to the demonstrated Medical Test Catalog, diagnostic-test selection, booking journey, appointment tracking, appointment-completion automation, report record and My Lab Reports.
- An actual external payment gateway or settlement integration is not claimed merely because the portal contains a Payment stage; it should be separately implemented and validated.
- Direct hospital/EMR/LIS integration, external messaging, advanced analytics, predictive clinical capabilities and AI clinical decision support are not claimed unless separately implemented and demonstrated.
- Autonomous medical diagnosis or treatment decisions are outside the ServiceNow workflow boundary.

11. FINAL ARCHITECTURE SUMMARY

The Medical Diagnosis Center architecture is a patient-facing ServiceNow diagnostic-service workflow. The central path is not a generic REQ/RITM approval workflow; it is the screenshot-supported journey from Medical Test Catalog through Book Now, Patient Details, Schedule, Payment and Done, followed by Appointment tracking, appointment completion, automatic Report creation and My Lab Reports access. This architecture should be used as the reference architecture for the remaining project documentation so that the problem statement, proposed solution, fit analysis, workflow, demonstration and implementation documents all describe the same actual system.